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Exploring point-of-interest data from social media for artificial surface validation with decision trees

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ABSTRACT

Artificial surfaces represent one of the key land cover types, and validation is an indispensable component of land cover mapping that ensures data quality. Traditionally, validation has been carried out by confronting the produced land cover map with reference data, which is collected through field surveys or image interpretation. However, this approach has limitations, including high costs in terms of money and time. Recently, geo-tagged photos from social media have been used as reference data. This procedure has lower costs, but the process of interpreting geo-tagged photos is still time-consuming. In fact, social media point of interest (POI) data, including geo-tagged photos, may contain useful textual information for land cover validation. However, this kind of special textual data has seldom been analysed or used to support land cover validation. This paper examines the potential of textual information from social media POIs as a new reference source to assist in artificial surface validation without photo recognition and proposes a validation framework using modified decision trees. First, POI datasets are classified semantically to divide POIs into the standard taxonomy of land cover maps. Then, a decision tree model is built and trained to classify POIs automatically. To eliminate the effects of spatial heterogeneity on POI classification, the shortest distances between each POI and both roads and villages serve as two factors in the modified decision tree model. Finally, a data transformation based on a majority vote algorithm is then performed to convert the classified points into raster form for the purposes of applying confusion matrix methods to the land cover map. Using Beijing as a study area, social media POIs from Sina Weibo were collected to validate artificial surfaces in GlobeLand30 in 2010. A classification accuracy of 80.68% was achieved through our modified decision tree method. Compared with a classification method without spatial heterogeneity, the accuracy is 10% greater. This result indicates that our modified decision tree method displays considerable skill in classifying POIs with high spatial heterogeneity. In addition, a high validation accuracy of 92.76% was achieved, which is relatively close to the official result of 86.7%. These preliminary results indicate that social media POI datasets are valuable ancillary data for land cover validation, and our proposed validation framework provides opportunities for land cover validation with low costs in terms of money and time.

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1. Introduction

Artificial surface is a key land cover type formed by human activities (Liang et al. 2015) and refers to continuous and discontinuous urban fabric, industrial, commercial and transport units, road and rail networks, dump sites and extraction sites, etc (Pons and Rullan 2014). It often serves as an important indicator of human impacts on terrestrial ecosystems (Kuang et al. 2016). Therefore, on one hand, artificial surface validation is an important part of the process of land cover validation. On the other hand, as a means of assessing the quality of information, artificial surface validation is assumed to justify the application of artificial surface products for various applications involving terrestrial ecosystems. In traditional methods, validation is carried out by confronting the produced land cover map with reference data, which include 'ground truth' data collected directly in the field at pre-selected sites or interpreted from remotely sensed images (Estima and Painho 2013a). These methods have limitations, including high costs in terms of money and time. Recently, geo-tagged photos from social media (such as Flickr) have been used as reference data for land cover validation (Foody and Boyd 2013). These geo-tagged photo-based methods have lower costs because of the cost-free nature of geo-tagged photos; however, they are still time-consuming due to the process of interpreting geo-tagged photos (Hou et al. 2015). The above advantages and disadvantages of geo-tagged photos have inspired other researchers to employ new geo-tagged social media data without using photo recognition techniques for more efficient land cover validation.

In fact, some social media point of interest (POI) data, including geo-tagged photos and other geo-tagged data, may contain useful textual information for land cover validation. Taking social media POIs as examples, the categories 'bank' and 'pasture park' correspond to the land cover types 'artificial surface' and 'grassland,' respectively. In addition, social media POI data also contain addresses and/or latitude-longitude information, which can correspond to the geographic locations of land cover reference data. Meanwhile, social media POIs are frequently updated by some social media providers and their users, which means their collection times can be related to the collection time of the land cover reference data. Therefore, exploring social media POIs without using photo recognition techniques for land cover validation may be possible.

Utilizing social media POIs requires extensive work to solve some fundamental questions. Although some categories of social media POIs are useful for land cover, there are still many categories that are not useful or ambiguous, such as the category 'entertainment places'. Therefore, how to properly clean and classify POI data to correspond to land cover types becomes an important issue to handle first. To date, some efforts have been made on this issue for classification of POIs based only on their categories and/or visual features in other applications involving land cover/use (Jiang et al. 2015; Hu et al. 2015; Leung and Newsam 2015). However, these efforts ignore the spatial heterogeneity of social media POI distributions. This heterogeneity can substantially influence both classification and validation accuracy because many POIs are located far apart and show a low density. For example, many POIs represented as 'Gas Station' or 'Bus Stop' are located in cultivated land, leading to incorrect classifications. Therefore, the first issue can be modified into how to clean and classify POIs, given their categories and spatial heterogeneity, to correspond to land cover types. Furthermore,

social media POIs are point data instead of spatial coverages, which is an important factor in land cover validation. Therefore, how to determine spatial coverage using classified POIs and their geographic locations becomes essential for land cover validation.

This paper attempts to explore the potential of social media POIs without using photo recognition techniques for land cover validation and proposes a validation framework using modified decision trees. First, POI datasets are classified semantically to divide POIs into the standard taxonomy of land cover maps. To solve the primary issue of spatial heterogeneity mentioned above, a modified decision tree model is built and trained to classify POIs automatically. Furthermore, a data transformation based on a majority vote algorithm with geographic locations and classification results from POIs is applied to transfer each point into a cell, which fits the spatial coverage and the resolution of the land cover map. Finally, the transformed POI data were used to validate areas with the artificial surface type in China's 30-meter global land cover datasets (GlobeLand30) for the year 2010.

2. Related studies

2.1. Social media data resources and applications

With the advent of Internet technology, social media data, such as geo-tagged social media texts, geo-tagged photos and geo-tagged POIs, have emerged, offering massive opportunities for scientific studies and relevant applications. Geo-tagged web texts and photos are usually generated by social media networks, and locations are captured via built-in GPS units in mobile devices. For example, users can add location information while posting tweets by accessing the geographic locations of their mobile devices.

Currently, social media data are widely used in much geo-location related research. For example, social media data on noise complaints on mobile apps can be used for city noise detection (Zheng et al. 2014). In terms of urban planning, social media check-in data have the potential to reflect intra-urban human mobility (Wu et al. 2014), and, together with tweets and POIs from Yahoo!, can be applied to analyse urban spatial structures (Huang and Wong 2016). In addition, urban areas of interest (AOIs) can be mined effectively using social media data such as geo-tagged photos (Hu et al. 2015). All of these studies have provided deep explorations of current social media data and motivated further research, proving that social media data have huge potential in filling the gaps in traditional data.

Furthermore, many attempts have also been made to use social media data in land cover and land use (LULC) analysis. For example, geo-located tweets were utilized for categorizing urban functional zones (Frias-Martinez and Frias-Martinez 2014). Leung and Newsam (2012) classified geo-tagged photos on Flickr for land-use classification. Shelton, Poorthuis, and Zook (2015) analysed social media data and obtained a complex understanding of socio-spatial relations. Feick and Robertson (2015) proposed an approach to analysing geo-tagged photos for exploration of urban places.

Along with the growing interest in analysing social media data in various domains, concerns about the limited investigations of the quality and efficiency of the data classification have been raised by researchers in many studies.

2.2. Social media data classification

Social media data classification is essential in geo-location relevant studies due to its large quantity and the lack of specific description labels. Many studies have proposed different approaches to various social media data classifications, such as OpenStreetMap (OSM) check-in data. For example, Estima and Painho (2013b) classified OpenStreetMap polygon data based on the semantic meaning of text attributes. Ali et al. (2016) defined appropriate and inappropriate classifications based on the environment factors associated with OSM data to exploit the characteristics of data classification. Frias-Martinez and Frias-Martinez (2014) applied spectral clustering to geo-located tweets to identify land use parcels. Generally, textual information, posting time and location, as well as environmental factors, are considered in the effective classification of social media data, and could also improve social media POI classification.

Social media POIs, along with coordinate information, brief tags and sometimes images or photos on Flickr, have typically been classified manually or automatically in previous studies. For example, Estima, Fonte, and Painho (2014) applied an expert classification of Flickr photos based on image content interpretation to agree with land cover nomenclature. Antoniou et al. (2016) extracted 1000 geo-tagged photos and chose seven volunteers to classify them to investigate the feasibility of geo-tagged photos as land cover input data. In these studies, these photos were classified manually, which tends to be subjective, to depend on the volunteers' recognition of data features and cannot employ large quantities of data, given the limitations imposed by the labour involved. Therefore, an automatic method is necessary for social media data classification. Zheng, Liu, and Hsieh (2013) detected areas which are related to air pollution using POIs, traffic flow, meteorology and road networks by applying artificial neural networks (ANNs) and a linear-chain conditional random field (CRF). Leung and Newsam (2012) explored tags in Flickr with semantic modelling to group different kinds of photos. Jiang et al. (2015) compared nine algorithms for classifying POIs from Yahoo! for the purpose of estimating urban employment size, and the best result showed an accuracy of 86.646% using the IBk algorithm. The reason why this instance-based model yielded a satisfactory result is that the distribution of different Yahoo! categories implied no clear dominance of some categories over others. Liu and Long (2015) applied a cellular automata (CA) model using OSM data and POIs as environmental factors to identify city parcels in China.

As discussed above, manual and automatic classification using only data attributes have been applied in land cover validation. Classification methods considering spatial heterogeneity have not yet been utilized to support land cover validation.

2.3. Social media data for land cover validation

Land cover validation is widely regarded as an essential part of map production, without which the map will have little value for all kinds of applications (Brovelli et al. 2015). It is usually done by experts who take each sample as 'ground truth' and compare it with a land cover map (Foody 2010) or evaluate classification accuracy by taking other land cover products as references. For example, Arsanjani, See, and Tayyebi (2016) compared the Globeland30 product with data from Coordination of Information on the

Environment (CORINE), Urban Atlas, OpenStreetMap and Amtliches Topographisch-Kartographisches InformationsSystem (ATKIS) covering parts of Germany to assess the class agreement. However, these approaches are limited by their high costs and are usually time-consuming. Moreover, some sites may not be categorized easily, as experts do not always agree with one another (Fonte et al. 2015).

With the development of information technology, social media data are becoming a new source of data for land cover validation. These data sources potentially contain land cover types and geographical coordinates, which are necessary to contribute constructively to land cover assessment (Liu et al. 2015). Currently, two social media data sources have been explored to validate land cover maps: geo-tagged web texts and geo-tagged photos. Geo-tagged text, such as tweets posted by users and other social media information published online, is an effective method for land cover validation. For example, Hou et al. (2015) collected textual information from websites using a deep web crawler to validate land cover maps. These web texts, which include location and time attributes, are posted by users who are willing to rent or sell their apartments, so it can potentially be included in the data collection process. Jenkins et al. (2016) derived semantic meaning from Twitter and Wikipedia to identify thematic characteristics of places in terms of their land use. Compared with remotely sensed images and field survey samples, these data collection methods have lower costs because the data employed are completely free. Although the studies considered the use of semantic meaning data in classification and validation, they ignored the spatial heterogeneity of the data distribution, which can influence the accuracy result of land cover validation.

Geo-tagged photos, which are published on Flickr, Geograph, Instagram and other websites, represent another data source for web-based validation. Photos are generated by users who take pictures of their environment and record their locations online. For example, Foody and Boyd (2013) used geo-tagged photos as reference data to validate the extent of tropical forest in West Africa. Researchers can collect data samples easily using online photos rather than interpreting remotely sensed images or taking field survey samples. The text information that accompanies these photos has also been considered as ancillary information for validation. For instance, Leung and Newsam (2015) extracted photos and images from a collection of texts from Flickr and Geograph and used them as training data in a supervised learning framework to support land cover classification. However, no previous studies have utilized geo-tagged photos for land cover validation based mainly on the textual information. Moreover, spatial heterogeneity is also not considered in these geo-tagged photo-based studies.

3. Methodology

3.1. Validation framework

The main goals of this study are to explore the accuracy of social media POIs for artificial surface validation considering spatial heterogeneity and to propose a validation framework that uses modified decision trees. The proposed framework consists of three key components, including POI classification, POI generalisation and artificial surface validation. The framework is shown in Figure 1.

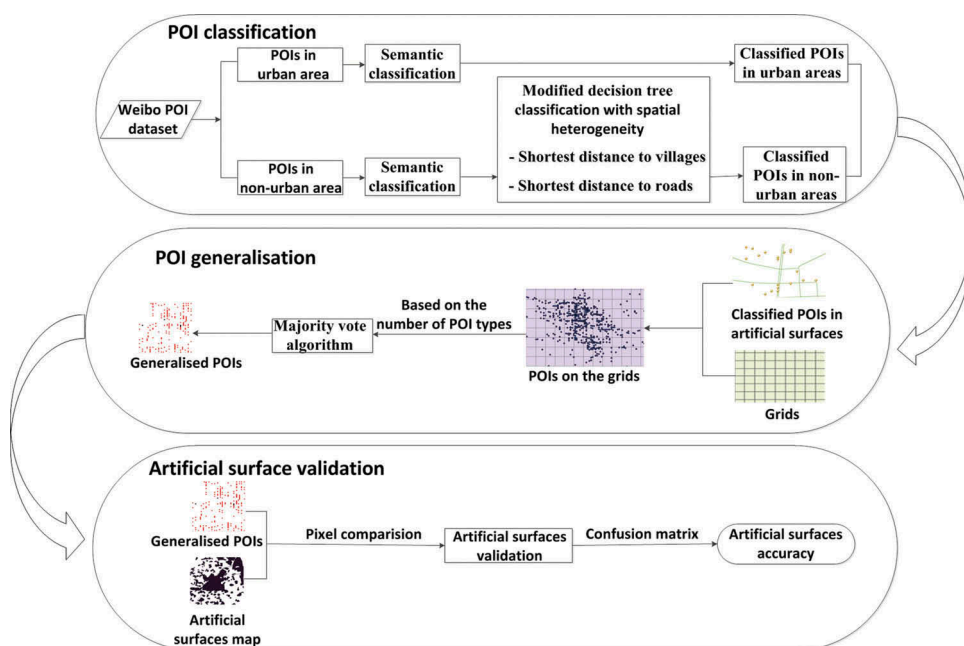


Figure 1. The overview of the framework.

- **POI classification.** It aims to accurately group POI data into artificial surfaces and non-artificial surfaces. First, the POI data are divided into urban and non-urban areas. Then, POI data from urban and non-urban areas are classified as artificial surfaces and other land cover categories in a semantic way, respectively, as in previous studies (Jiang et al. 2015; Hu et al. 2015; Leung and Newsam 2015). In fact, POI data in non-urban areas are sparse. Thus, it presents high spatial heterogeneity, which can lead to unsatisfactory results in the classification. To solve this problem, the shortest distances between each POI and both roads and villages serve as two factors in the modified decision tree model for reclassifying POI data in non-urban areas.
- **POI generalisation.** In this component, the majority vote algorithm based on the number of POI types is used to transform the classified POI data into raster form and identify land cover type for the purposes of validation.
- **Artificial surface validation.** It aims to estimate the accuracy of land cover maps using the generalised POI data. In this component, a common pixel comparison method and confusion matrix methods are applied to estimate the accuracy.

3.2. Artificial surfaces classification with modified decision trees

3.2.1. POI distribution analysis

While previous work classified POIs based only on their semantic attributes, our study takes POI spatial heterogeneity into consideration to identify POI types. POI Spatial heterogeneity indicates the uneven distribution of each POI category types within an area. In general, urban areas display an extremely high density of POIs which are usually located in artificial surfaces (POIs in Figure 2(b)), while POIs in non-urban areas show

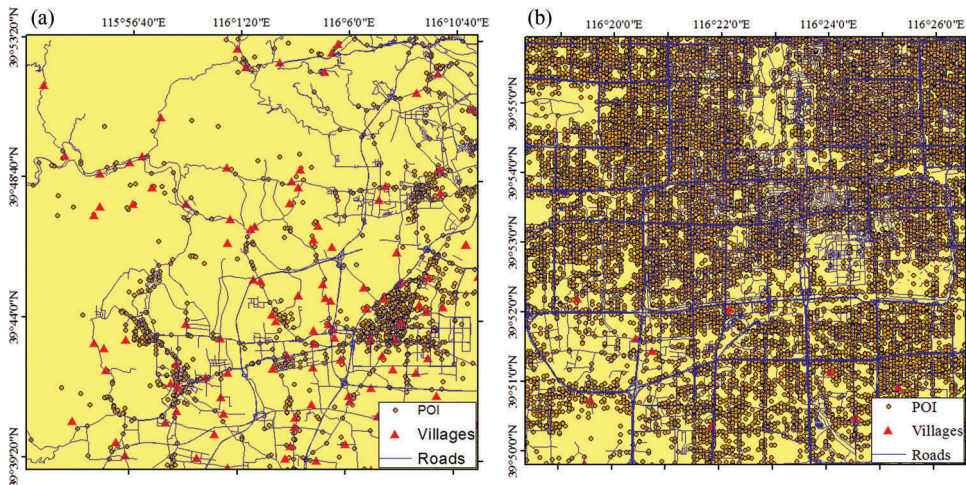


Figure 2. The distributions of POI in non-urban areas and urban areas. (a) POI, village and road distributions in non-urban areas. (b) POI, village and road distributions in urban areas.

high spatial heterogeneity: the distribution of them is relatively sparse and these POIs distributes in several land cover types (POIs in Figure 2(a)).

To reduce the effect of spatial heterogeneity on POI classification, spatial context factors based on position and proximity which corresponding to artificial surfaces should be involved. Although cultivated land, forest and grassland are dominated land cover types in non-urban areas, villages usually distribute in artificial surfaces indicating that POIs around villages have more probabilities to belong to the same type. Furthermore, shops, gas station and other public facilities near the roads which are categorised in artificial surfaces usually indicate that POIs in these regions are more likely to be located in artificial surfaces. In Figure 2(a), areas in which a few POIs are near the villages can be classified as artificial surfaces. Besides, POIs located beside the road become the indicator of artificial surfaces. Therefore, POIs can be easily classified considering these two factors. In contrast, POIs in urban area do not display heterogeneously. As is shown in Figure 2(b), POIs in the urban areas are impervious to the distance between villages and road. The reason is that artificial surfaces are the dominant land cover type in urban areas. And the large number of POIs convince the reliability of types in urban areas.

Based on the analysis above, this paper considers two factors, the shortest distance from POIs to roads and villages with the purpose of determining potential rules for distinguishing POI types and their distributions effectively, using a modified decision tree to obtain an optimal classifier. As one of the effective decision tree algorithms which specializes in finding potential relationships between numerous inputs and the output POI types, the C5.0 algorithm (Quinlan 1998) is applied, which can assign each input parameter a proper weight and a meaningful cost so that the agreement between the real and predicted POI types can be improved.

3.2.2. Modified decision trees with adaptive boosting and a cost matrix

Decision trees, a supervised classification method in machine learning, provide insight and understanding into the predictive construction of data. Each decision tree algorithm

relies on different mathematical principles to build classification rules based on the input parameters. As the proposed decision tree algorithm, C5.0, uses entropy to measure purity which indicates the classes contained in the data samples. The minimum value 0 indicates the data are totally homogeneous, while the maximum value 1 represents complete heterogeneity of the data sample.

In our study, entropy is used to build the land cover classifier. Value 0 represents that POI samples only contain the type of artificial surfaces or non-artificial surfaces while other values indicate that POI samples contains both two types. We define a POI dataset S which consists of n POIs. The term c refers to the number of different types of POIs in which each type has different values $C_i (i = 1, 2, 3, \dots, c)$. The formula that defines the entropy is shown as Equation (1):

$$E(S) = \sum_{i=1}^c -p_i \log_2(p_i) \quad (1)$$

where $E(S)$ indicates the entropy of POI dataset S . p_i represents the percentage of POIs falling into land cover type C_i . p_i can be calculated using Equation (2):

$$p_i = \frac{n_i}{|S|} \quad (2)$$

where $|S|$ denotes the number of POI data n .

Land cover classification is proposed based on spatial context factors: the shortest distance between each POI and a village, and the shortest distance between each POI and a road. To decide which factor to be classified, C5.0 uses entropy to calculate the information gain, which indicates the change in homogeneity. The value of the information gain is determined as Equation (3):

$$G(A) = E(S_1) - E(S_2) \quad (3)$$

where A represents the attribute of the POI dataset. $G(A)$ indicates the information gain given the attribute A . S_1 denotes the number of POIs before classification and S_2 denotes the number of POIs in the subset samples. To calculate $E(S_2)$, each classified POI entropy should be considered and summed up with a weight. It is defined as Equation (4):

$$E(S_2) = \sum_{j=1}^n w_i \sum_{i=1}^c p_{ij} \log_2(p_{ij}) \quad (4)$$

where j denotes the number of land cover types in subset S_2 . w_i denotes the proportion of the number of POIs to all the samples in the dataset. p_{ij} denotes POIs whose value is in class C_i . In the end, land cover classifier is established based on the information gain determined by calculating entropy of each POI samples.

Although this decision tree model, which is based on entropy measurement, is quite reasonable, we may have a better option to choose which type to classify with higher accuracy at the cost of worse results for other types. Furthermore, due to the relatively low accuracy of each original classifier, many weak decision tree classifiers can be trained to combine a stronger one. One of the most effective ways to solve the problems above involves adding adaptive boosting and establishing a cost matrix based on the target type.

Adaptive boosting is based on iteratively learning a large proportion of POI samples which are difficult to classify by assigning a weight to each sample. The proposed boosting iteration algorithm is set as follows:

With n POI samples are included in dataset S , the aim of boosting is to construct T land cover classifiers, where T also indicates the number of trials in the boosting algorithm. C^t denotes the land cover classifier in trial t and ω_i^t indicates the weight of POI sample i in trial t . In addition, an indicator function is defined as in Equation (5):

$$y_i = \begin{cases} 1, & \text{POI sample is classified correctly} \\ 0, & \text{POI sample is classified incorrectly} \end{cases} \quad (5)$$

The first step in boosting is to initialize classifier number T . we assume that the original number of land cover classifiers is 1 and the weight of the first classifier is $\omega_i^1 = 1/n$. For each trial, the land cover classifier which is modelled with decision trees is trained to minimize the weight error of POI samples. The weighted error function is defined as Equation (6):

$$\epsilon^t = \sum_{i=0}^0 \omega_i^t I(C^t(i) \neq y_i) \quad (6)$$

The weight α^t of each land cover classifier is then calculated based on the weight error of POI samples ϵ^t as Equation (7):

$$\alpha^t = \frac{1}{2} \ln \left(\frac{1 - \epsilon^t}{\epsilon^t} \right) \quad (7)$$

According to the weight α^t modified by the weight error of POI samples ϵ^t , the updating weight ω_i^t is then given by Equation (8):

$$\omega_i^{t+1} = \frac{\omega_i^t}{Z^t} \times \begin{cases} e^{-\alpha^t}, & \text{POI sample is classified correctly} \\ e^{\alpha^t}, & \text{POI sample is classified incorrectly} \end{cases} \quad (8)$$

where Z^t represents the normalized factor of updating weight ω_i^t . e indicates the universal constant, adjusting the updating weight ω_i^t based on the weight α^t . The final step is assembling every land cover classifier with adjustable weight α^t . The final land cover classification model C^T is given by Equation (9):

$$C^T = \sum_{n=1}^n \omega_i^t C^t \quad (9)$$

Cost matrix is set as the misclassification cost of each land cover type. The higher the cost is, the greater the cost of land cover misclassification becomes. Compared with the equal treatment of misclassification errors, land cover type with higher costs can be given greater weights to improve the final land cover classification accuracy. The construction of the cost matrix is shown as follows:

Classified as non-artificial surfaces Classified as artificial surfaces

$$\begin{matrix} \text{Non-artificial surfaces} \\ \text{Artificial surfaces} \end{matrix} \begin{bmatrix} 0 & m \\ n & 0 \end{bmatrix}$$

According to the matrix, there is no cost assigned when a POI sample is classified correctly. However, those POI samples which are misclassified as non-artificial surfaces have a cost of n , while others which are misclassified as artificial surfaces have a cost of m .

3.2.3. Classification features considering spatial heterogeneity

While the distributions of POIs tend to show a high density with mainly artificial surfaces type in urban areas, POIs in non-urban areas are often have a low density with various of land cover types, leading to the problem of spatial heterogeneity. Therefore, how to perform correct classifications, given the spatial heterogeneity of POIs, becomes of vital importance in building and training POI-classification decision tree models.

Considering that POIs in non-urban areas are usually located in places which are near roads or villages, given their easy accessibility for people, the decision tree classifier can be trained with the parameters of the shortest distance between each POI and a village and the shortest distance between each POI and a road. Both villages and roads datasets are extracted from OSM downloaded from the <http://www.openstreetmap.org/>. The shortest distance between POIs and villages is calculated based on the Euclidean metric. For the shortest distance between POIs and roads, the calculation is defined as the vertical distance between them.

The calculation results of these two parameters are shown in Figure 3. Most of the POIs are located far from the nearest village (within 2000 m), while POIs are far nearer to roads, as many POI locations occur on roadsides. Several abnormal values can be seen with obviously larger values. For those POIs which are located far away from villages or

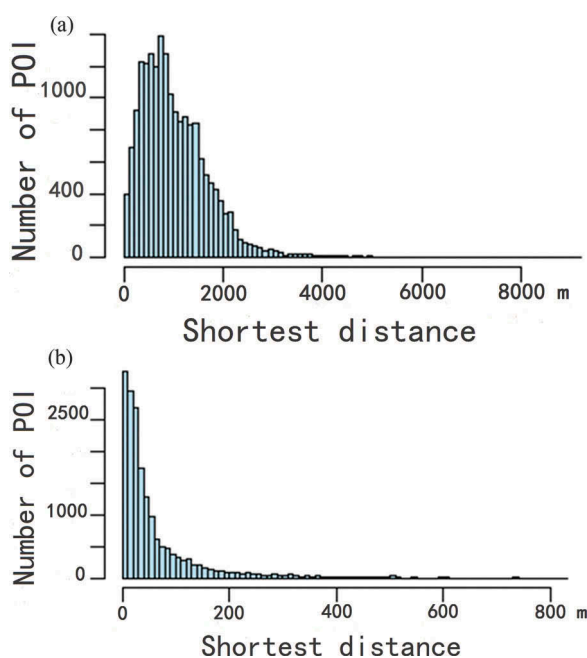


Figure 3. Distribution of two calculated parameters. (a) Shortest distance to villages. (b) Shortest distance to roads.

roads, the land cover types they belong to are more likely to be cultivated land. That is probably because several public facilities are needed in these areas, which may lead to the misclassification of POIs.

To avoid the situation that the trained decision tree model performs under fitting or over fitting, in one region, and shows a very low accuracy in other areas, the boosting algorithm that we discussed above can be used to train multiple land cover classifiers and combine them into a stronger one. First, the number of land cover classifiers T to be trained need to be determined. To choose the proper number T , a number 1 to 10 is set as the number of land cover classifiers separately and the error rates are compared among them. The reason that classifier number T is not set to values greater than 10 is that the number T is strongly related to the classifier size: the more classifiers we set, the smaller the classifier size is. When classifier number T is greater than 10, the classifier size is not large enough to build the final land cover classification model.

Defining a cost matrix is another way to improve the decision tree model. Considering that our target classification type is artificial surfaces, the cost of misclassifying one POI as an artificial surface (cost m) should be much higher than that associated with misclassification as a non-artificial surface (cost n). Therefore, in the cost matrix we established, the cost m should be set greater than n . Noting that m and n are only the relative weights of misclassifying as artificial surfaces and non-artificial surfaces, there is no differences between $m = 2, n = 1$ and $m = 4, n = 2$. To obtain the proper weights m and n , we set $m = 1, n = 1$ originally and gradually raise the value of m in order to set more costs in misclassifying as artificial surfaces. The proper weights depend on the accuracy of decision tree classification model.

3.3. POI data generalisation based on the majority vote algorithm

To validate the artificial surface map, POI datasets should be transferred to a grid format. Although a rasterization process should be done to fit land cover maps, the final type of each grid can be uncertain because POIs with different types may exist in the same grid. Therefore, POIs should be generalised to one type based on the numbers of each type.

First, questions about how these contributions are merged together should be addressed. We decided to employ a grid with a resolution of 30 m $\times$ 30 m because it can exactly match the resolution of the GlobeLand30 data product so that all the contributions within the same cell can be evaluated together without ruining the original morphological patterns.

Then, the value of each grid should be determined based on the classified POI type. On one hand, if all POIs falling into one grid belong to the same land cover type, the grid value is that POI type. On the other hand, if one grid contains different types of POIs, the majority vote algorithm (Boyer and Moore 1991) is applied to determine the final value of each grid. The majority vote algorithm was originally developed to determine which candidate has received a majority of the votes cast in an election and includes two parts: the pairing phase and the counting phase. For the POI generalisation problem in this study, the first part is to calculate the number of POIs of each type. A counting variable k which is initially set as 0 and a POI type indicator v are established. The value of indicator v is set as the current calculating POI types. Every time the currently calculated POI type equals v , counting variable k is incremented by 1.

Otherwise, the value of counting variable k decreases by 1 until k equals 0. In that case, the type of the currently calculated POI becomes the value of the grid. The second part is to calculate the number of POIs of the selected land cover type when the value of the type in each grid is settled. The final land cover type of each grid can be finally determined if the number of selected POIs is more than 1/2 of the total number in each grid.

After the data generalisation process is done, the confusion matrix is used as a performance indicator to check the proposed model's quality and finally the accuracy of the results of the artificial surface validation.

4. Experiments and results

4.1. Study area and experimental data

The study area examined in this paper is Beijing, the capital of China. Containing a total area of 16,410.54 km², Beijing includes various landscapes, including cultivated land, forest, grassland, wetland, water bodies and artificial cover, presenting challenges in that POIs are distributed heterogeneously over these areas (Figure 4). Moreover, as one of the biggest cities in China, it has received many POI contributions, so we can obtain relatively more accurate results.

In this study, POIs are collected using Sina API. Sina Weibo provides the location search interface 'place/nearby/pois' which can search the POI information based on the location and posting time. Considering the temporal limitations of the API, we write a crawler in Python which allowed data to be gained 1000 times per hour. The obtained messages contain attributes including the location name, POI type and coordinates. Because the coordinates Weibo provides have been coded, decoding work must be done to correct the geographical locations. The coordinates are transferred into World Geodetic System 1984 (WGS84), the same geographic coordinate system as the GlobeLand30 map.

To improve the quality of POIs, POIs with ineffective categories are excluded. For example, many POIs are labelled as 'unclassified category' indicating places where are not classified with POI types. These POIs are considered as data noise because they could be located in anywhere. Based on the above data pre-processing, a total of 141,745 POIs remain to be employed for land cover classification.

The GlobeLand30 data product contains ten land-cover types with only one level of classification (Chen et al. 2015; Chen, Ban, and Li 2014). With its resolution of 30 m × 30 m, the data product includes land-cover maps for both 2000 and 2010. In this paper, we validate the GlobeLand30 map for the year 2010. The nomenclature used in GlobeLand30 is shown in Table 1.

4.2. Data classification

To classify POIs considering both semantic meaning and spatial heterogeneity and to illustrate the potential effects of utilizing the modified decision tree model, the classification experiment is separated into two parts: the first part involves semantic

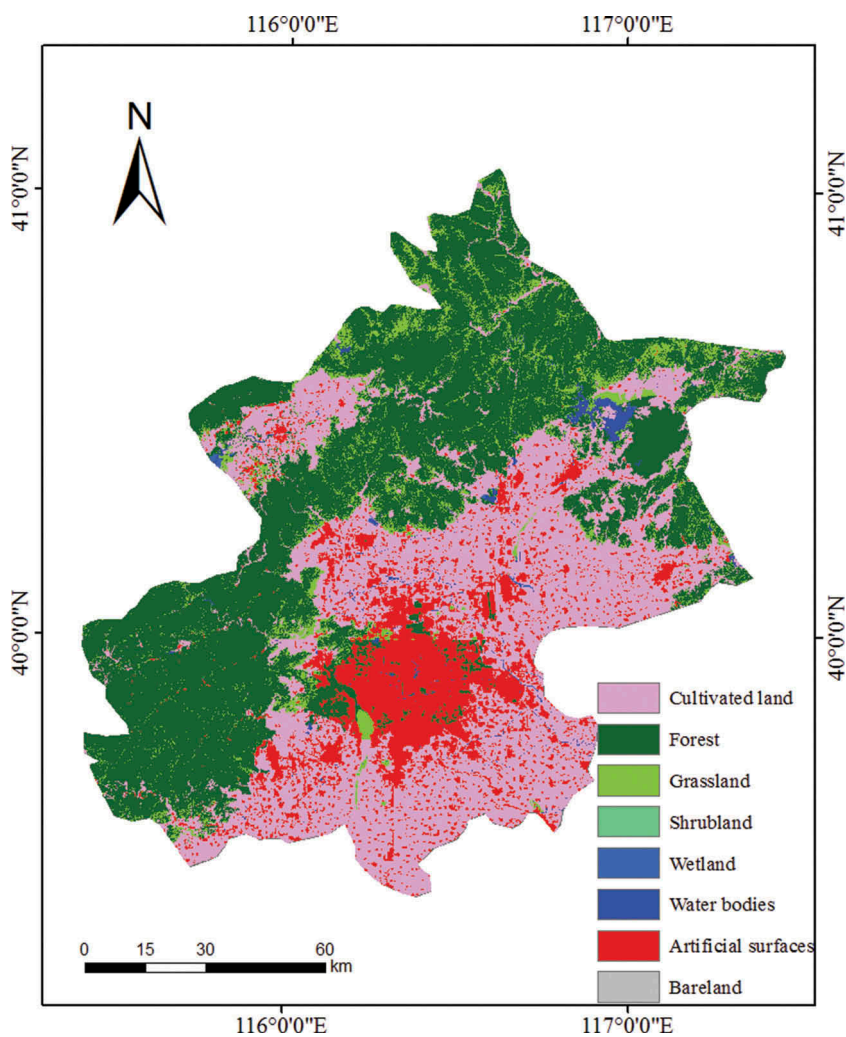


Figure 4. GlobeLand30 map for 2010 in the area of Beijing.

Table 1. GlobeLand30 nomenclature.

Code	Type
10	Cultivated land
20	Forest
30	Grassland
40	Shrubland
50	Wetland
60	Water bodies
70	Tundra
80	Artificial surfaces
90	Bareland
100	Permanent snow and ice

classification without addressing spatial heterogeneity, and the second part includes spatial heterogeneity-based POI classification after the semantic filtering. In the end, we will show the classification results of these two experiments.

4.2.1. Experiment 1: POI classification without considering spatial heterogeneity

For POI semantic classification, POIs are extracted based on the offered types of POIs to be matched with GlobeLand30 categories. In fact, despite the various names of POIs, they are grouped into different types such as ‘Chinese Restaurant’ and ‘Supermarket’ (Table 2), and three official levels of categories are offered by Sina Weibo (the type attribute in Table 2 is level three). We use level three to apply semantic classification because level one and level two can sometimes contain more than one GlobeLand30 land cover type. Part of the semantic classification of level three is shown in Table 3.

As shown in Table 3, there are five equivalent categories in the GlobeLand30 map that correspond to POI types, including grassland, water bodies, forest, cultivated land and artificial surfaces. Apart from that, several types are said to be unclassified because of unclear semantic descriptions in the texts. Some definitions in the GlobeLand30 nomenclature need to be explained because special cases in the categories may have a great influence on the POI classification. (1) Cultivated land includes not only lands used for agriculture but also for horticulture and gardens. (2) Artificial surfaces are lands modified by human activities. Towns with large amounts of vegetation cover whose underlying surfaces are still impervious belong to artificial surfaces, while the green spaces in a city, such as parks and golf courses, are not. (3) Grassland includes areas of natural grass with over 20% vegetation cover. Livestock production areas are also included in this category. On the other hand, green areas in the city which are extracted from artificial surfaces belong to grassland. That is the reason why the POI types ‘golf course’, ‘garden’ and ‘farm land’ are excluded from the artificial surfaces and are

Table 2. Part of the POI dataset from Weibo.

Name	Types	Latitude	Longitude
Laodu restaurant	Chinese restaurant	116°23'17" E	39°46'30" N
Fuhailuyuan supermarket	Supermarket	116°23'12" E	39°46'30" N
Jade park	Park	116°19'19" E	39°46'3" N
Wanliu golf club	Golf	116°25'37" E	39°57'45" N
Qinghe	Bathing beach	116°22'1" E	40°1'43" N

Table 3. Grouping POIs according to different GlobeLand30 category labels.

GlobeLand30 types	POI types
Artificial surfaces	automobile service-related; gas station; automobile service-related; law office; gas station; Beijing cuisine; Accountants office; SINOPEC; Hubei cuisine; assessment office; CNPC; Dongbei cuisine; audit office; Shell; Yunnan-Guizhou cuisine; Authentication office; Mobile Northwest cuisine; Patent office; Caltexold brand; Personnel Market; Huarun; chafing dish restaurants; waterworks; business hall; Eastern Bloc; local dishes restaurant; electric business hall; PetroChina YPC acetyls; Seafood Pavilion; Beauty Salon; vegetable dishes
Grassland	golf-related; botanical garden; park square; golf course; pasture park; golf driving range; farm
Water bodies	bathing beach; transit-port; passenger port; fishing garden bridge; ferry of vehicle; water activity centre; fishing ground; passenger ferry; port terminals; beach
Forest	forest farm
Cultivated land	vegetable base; fruit base; flower and nursery
Unclassified types	Cemetery; the base of poultry raising; famous scenery; cemetery; information on placename and address; world heritage; ski tourism; traffic place-name; national scenic spot; horse racing ground island ring-name; provincial scenic spot; entertainment places; public facilities; traffic service-related; the place of vacation and convalescent; emergency shelter; emergency refuge space; the base of Faming, Forestry, Animal Husbandry and Fishery; leisure place; picking garden

Table 4. Number and proportion of each POI type.

Code	Type	Number of POIs	Proportion (%)
30	Grassland	1289	0.91
60	Water bodies	47	0.03
80	Artificial surfaces	140,022	98.78
0	Unclassified	387	0.27
	Total	141,745	100.00

grouped as grasslands, while ‘vegetable garden’ and ‘fruit garden’ belong to the agriculture land use type.

In the study area of Beijing, semantically classified POIs are calculated and the number and proportion of each type of POI is shown in Table 4. Only ‘Grassland’, ‘Water bodies’ and ‘Artificial surfaces’ are matched by the POI types, which indicates that not all the POI types occur in the Beijing area. Among the POI datasets, the majority of the POIs belong to ‘Artificial surfaces’ and make up 98.78% of the total number of observations, while POIs that belong to the ‘Grassland’ and ‘Water bodies’ types only make up 0.91% and 0.03%, respectively. The remaining parts of the datasets include unclassified types which occupy 0.27% of the datasets. For the purposes of extracting POI data which correspond to artificial surfaces semantically, other types of POIs are classified as non-artificial surfaces. Finally, the POIs are used to evaluate the semantic classification accuracy. To obtain the land cover type of each POI for classification accuracy evaluation, those POIs are overlaid with the image from the year 2010 in Google Earth, labelled with land cover types in the traditional way, and used as reference data. To compare the accuracy of the classification results with those in the next chapter, in which the POIs in non-urban areas are classified based on spatial heterogeneity, we divide the Beijing area into two parts, the urban areas where POIs are usually clustered regularly and the non-urban areas where POIs are often distributed with a low density. We use housing estate sources as reference data to identify the boundary between these two regions. 18,925 pieces of data are labelled non-urban areas and occupy 13.52% of the total dataset. The accuracy of the results in non-urban areas in experiment 1 is shown in Table 5.

4.2.2. Experiment 2: POI classification considering spatial heterogeneity

To apply spatial heterogeneity based semantic classification, all the data from the urban area is used for validation in the next step, and two parameters, the distance between each POI and the nearest village, as well as the shortest distances between POIs and roads, are considered to determine whether the POIs truly belong to artificial surfaces in non-urban areas.

A modified decision tree algorithm, C5.0, is applied, using the two parameters we discussed above as model input. We divide the sample data into two groups. The first

Table 5. The classification result without consideration of spatial heterogeneity in non-urban areas.

	Predicted non-artificial surfaces	Predicted artificial surfaces	Total number
Artificial surfaces	263	4276	4539
Non-artificial surfaces	1362	13,024	14,386
Total number	1625	17,300	18,925
Accuracy (%)			70.214

Table 6. Data samples with the correct rates used for C5.0 classification.

	Training data	Testing data
POI number	10,000	8925
Proportion (%)	52.84	47.16
Correct rate (%)	70.23	70.24

group is used to build the decision tree model and the second group is utilized for prediction. Instead of splitting the data spatially, we select POIs at random and extract 10,000 of them as training data with the remaining part being used as testing data. Considering the potential problem that the correct rates in these two groups show significant differences from each other, simple calculations are performed to guarantee the similarity and accuracy of the two groups. The result shows that data which are correctly labelled make up 70.23% of the total dataset in the former samples and the corresponding percentage in the latter samples is 70.24%, which means that the split is satisfactory and is also consistent with the semantic classification result in experiment 1 (Table 6).

To reduce the number of false positives and improve the final accuracy, adaptive boosting is applied in this model. We set the number of trials to values between 1 and 10 times, respectively, to restrain the size of the decision tree and adjust the weighting of the decision trees in each trial. After the weighting has been adjusted, the accuracy in every trial is shown in Table 7. With an increase in the number of trials, the error rate of the boosting algorithm rises from 25.20% to approximately 30% for 6 trials, followed by a tendency to decrease, with errors of 20.40% in the final trial. As a result, we take 10 as the value of the boosting parameter, considering its relatively high accuracy.

A cost matrix is also added to the C5.0 algorithm to modify the decision tree model. As we aim to extract the POI data that correctly indicates artificial surfaces, the POIs which are supposed to represent the non-artificial cover should be given a greater 'penalty' than other mistakes. It should be noted that the classification accuracy can be improved with much higher cost m , but it may be also overfitting with training data. As is shown in Table 8, cost m is set as 1 to 5 with 10 trials in boosting algorithm, and the classification accuracy of training data and testing data can be calculated separately. When cost m is set as 1, 3 and 5, training data accuracy is 86.10%, 91.10% and 98.40%. However, the corresponding testing data accuracy is only 76.45%, 73.85% and 70.26%

Table 7. The decision tree size and the errors associated with each boosting trial using C5.0.

Trial	Decision tree size	Error (%)
0	5	25.20
1	2	24.70
2	4	23.50
3	4	24.50
4	5	36.90
5	4	24.60
6	6	36.80
7	2	23.40
8	2	24.90
9	3	22.20
10		20.40

Table 8. The classification result with different cost m for the training data and testing data.

Cost m	Cost n	Training data accuracy (%)	Testing data accuracy (%)
1	1	86.10	76.45
2	1	76.60	76.53
3	1	91.10	73.85
4	1	80.66	80.62
5	1	98.40	70.26

which means that the decision tree classification models have been over fitted. When cost m is set as 2, training data accuracy and testing data accuracy are 76.60% and 76.53% separately, which are relatively low accuracy. When cost m is set as 4, accuracy of both training data and testing data are approximately 80%. Compared with the accuracy when cost m is set as 2, model with cost m as 4 apparently shows a much better result. Noting that the model with cost m as 5 shows an obvious over fitted condition than others, m with cost which is bigger than 5 easily leads to overfitting. Therefore, we assume that the cost of m is four times higher than the cost of n .

The final results for the training and test data sets shown in Table 9 and Table 10 indicate that accuracy is improved by approximately 10% by applying the modified C5.0 decision tree model for both the training and test datasets compared with the corresponding accuracy from experiment 1 (70.214%). Although the number of classified POIs that are used for validation is much less than before, the higher quality data can be effectively used for land cover validation.

While the classification in non-urban regions is shown above, the POI in urban areas is still grouped in using the semantic method we proposed in the experiment 1. The final classification of artificial surface POIs and non-artificial surface POIs is displayed in Figure 5.

4.3. Globeland30 map validation

Because the Globeland30 map is produced in raster format with a resolution of 30 m $\times$ 30 m, the classified POI data are transferred onto the same grid. The current POI data are assigned to 66,058 grids. The number of each type in each grid is calculated

Table 9. The classification result considering spatial heterogeneity for the training data.

Training data	Predicted non-artificial surfaces	Predicted artificial surfaces	Total number
Artificial surfaces	321	1856	2177
Non-artificial surfaces	82	7741	7823
Total number	403	9597	10,000
Accuracy (%)			80.62

Table 10. The classification result considering spatial heterogeneity for the testing data.

Testing data	Predicted non-artificial surfaces	Predicted artificial surfaces	Total number
Artificial surfaces	631	1268	1899
Non-artificial surfaces	456	6570	7026
Total number	1087	7838	8925
Accuracy (%)			80.68

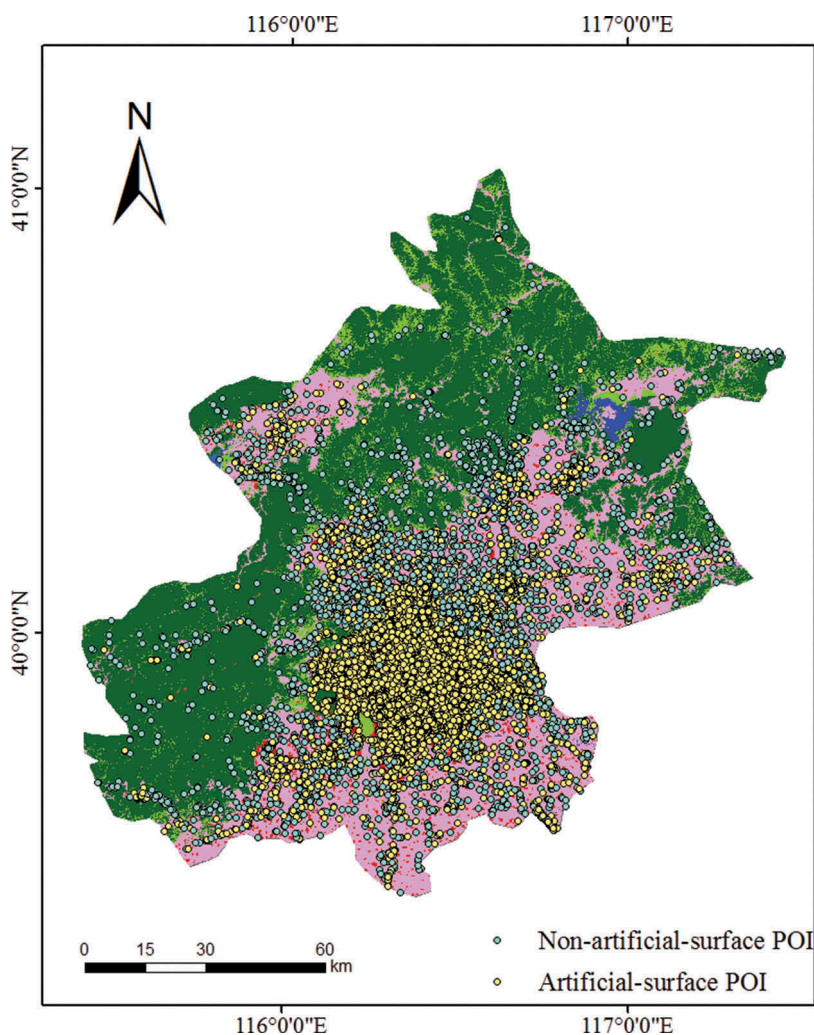


Figure 5. The final classification of POIs in Beijing.

and the land cover type is determined by applying the majority vote algorithm to the cells that contain both artificial and non-artificial POI types. To assess the validation accuracy of GlobeLand30 and evaluate the usability of the POI datasets, the GlobeLand30 data were reclassified to group the land cover types into two groups: artificial surfaces and non-artificial surfaces. Finally, confusion matrix calculation is utilized to calculate validation accuracy. Figure 6 shows the predicted artificial surfaces in raster format, and Figure 7 shows the reclassified result in GlobeLand30 map.

The final result of the artificial surface validation is shown in Table 11. The study area shows a high accuracy of 91.37% in total datasets. Compared with the accuracy rates reported in other papers (Estima and Painho 2015), social media POI data such as that supplied by Weibo has better usability than other kinds of datasets such as OSM. In terms of the accuracy of artificial surfaces, it indicates that 59,096 grids of a total of

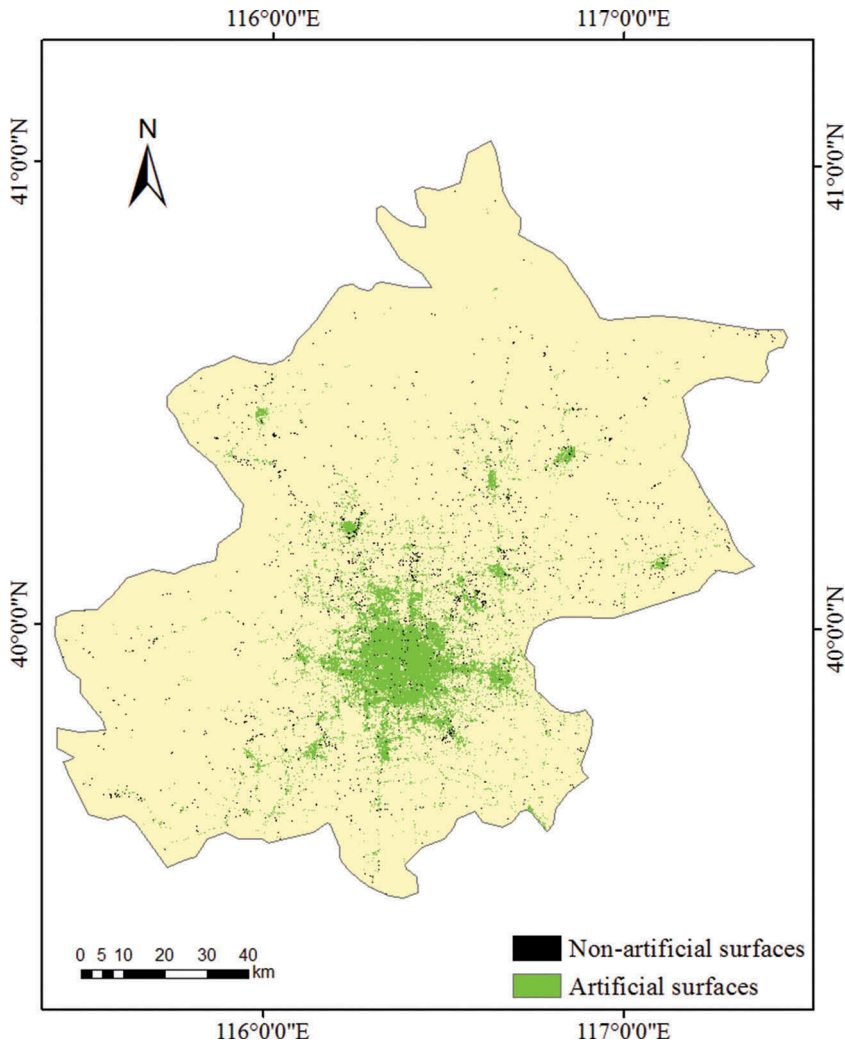


Figure 6. Distribution of POI artificial surfaces in 30 m resolution format.

66,058 grids which are labelled artificial surfaces are classified correctly, resulting in an accuracy rate of 92.76%.

Then, we do a detailed analysis based on the GlobeLand30 map (Table 12). The grids in urban areas and non-urban areas are analysed separately. Urban areas show a high rate of accuracy, although no further spatial heterogeneity-based classification process is employed. 503 and 794 grids which are classified as artificial surface are located in cultivated land and forest based on the GlobeLand30 map, respectively. Another 143 and 156 grids, which belong to grassland and water bodies, respectively, according to GlobeLand30, are not placed in the correct locations. Compared with urban regions, the consequence of non-urban area is much complicated. Apart from some POIs which are located in forest, grassland, wetland and water bodies on the GlobeLand30 map, most of the incorrectly classified grids are assigned a value corresponding to cultivated land. There are many reasons which may explain the occurrence of incorrect land cover types

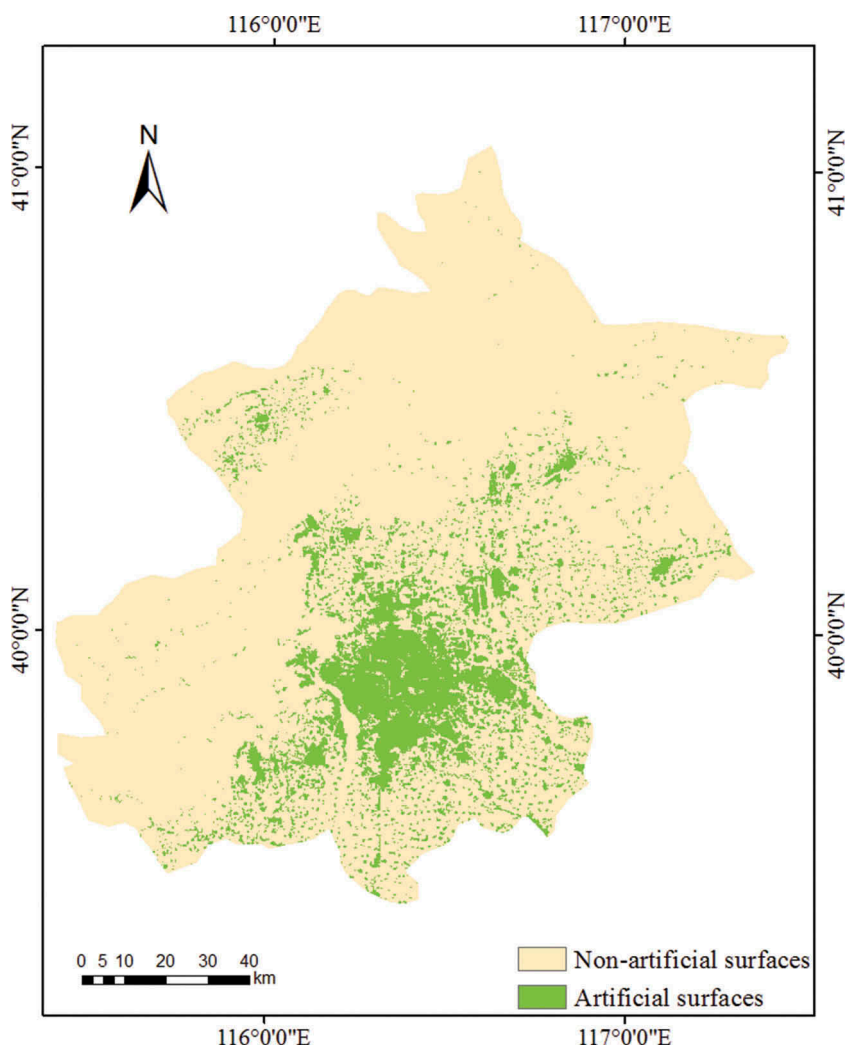


Figure 7. The reclassified result in GlobeLand30 map.

Table 11. Confusion matrix validation.

	Non-artificial surfaces in POIs	Artificial surfaces in POIs	Total number
Non-artificial surfaces in GlobeLand30	1259	4612	5871
Artificial surfaces in GlobeLand30	1091	59,096	60,187
Total number	2350	63,708	66,058
Accuracy (%)		92.76	91.37

in GlobeLand30. One is that some artificial parcels were missed during the production of the GlobeLand30 map in 2010, because a ‘minimum spot’ standard is used to perform the accuracy control of the GlobeLand30 map. When the land cover type is larger than or equal to the size of the spot, it should go through the quality control process in order to guarantee the quality of the land cover type data. The minimum spot varies from type to type where the standards for artificial surfaces are defined as follows: (1) non-urban

Table 12. Final results for the Beijing area.

	GlobeLand30 types	Classified as artificial surfaces
Urban areas	Cultivated land	503
	Forest	794
	Grassland	143
	Wetland	0
	Water bodies	159
	Artificial surfaces	51,388
Non-urban areas	Cultivated land	2648
	Forest	127
	Grassland	179
	Wetland	6
	Water bodies	31
	Artificial surfaces	7946
	Total	63,708

fringe zone, 8×8 pixels; (2) scattered cluster artificial cover, 4×4 pixels; (3) roads and residential areas distributed along roads, 5-pixel width. It indicates that any parcels whose grid sizes are below the minimum spot standard may be ignored.

In order to compare the time cost and accuracy of classification between geo-tagged texts and photos, 1543 geo-tagged photos are collected from Flickr in Beijing randomly. All the photos are interpreted respectively by two volunteers with approximately 1.5 h. Classification result shows that 872 photos are classified as artificial surfaces while 434 photos belong to non-artificial surfaces. The remaining 237 photos stay unclassified because landscape cannot be identified in these photos such as self-portrait. Therefore, a total number of 1306 classified photos are assessed and result shows that overall classification accuracy of photos is only 59.04% and the producer's accuracy of artificial surfaces is 69.01%. This result is similar to the average producer's accuracy of volunteers ranging from 52% to 62% proposing by Foody et al. (2013). In contrast, classified POIs can be obtained with merely a few second using the proposed classification method in this study. Classification accuracy with around 80% in Table 9 and Table 10 also indicates that POIs with textual information are relatively reliable to be applied in land cover validation.

5. Discussions

Our experiments confirm that the textual information from social media POI datasets are valuable in artificial-surface validation with low costs in terms of money and time, and the modified decision tree model is effective in addressing spatial heterogeneities in POI classification.

Our findings on using textual information from social media datasets for land cover validation are certainly similar to those of several existing approaches using geo-tagged social media datasets to support land cover validation (Estima and Painho 2013a; Foody and Boyd 2013). However, we used textual information related to artificial surfaces, rather than geo-tagged photos, as sample data for land cover validation, while image interpretation is usually used in traditional approaches. One challenge in dealing with textual information classification is the low density of POIs distributed over non-urban areas, which may lead to POI misclassification. To meet this challenge, we utilized a modified decision tree considering the shortest distance between POIs and villages, as

well as the shortest distance between POIs and roads, to classify POIs when POIs are distributed with low density, while most of the classification process only considered the semantic meaning of textual information.

Although the proposed approach helps identify an alternative data source and an effective classification method in terms of land cover validation, the traditional data sources and approaches for land cover validation are still valid and necessary. In fact, they complement each other; social media POIs and the proposed approach can be used to provide a good balance between efficiency and accuracy of land cover validation. For instance, social media POIs can be considered as an ancillary data source to help traditional approaches interpret images, while traditional data sources are also necessary for supporting land cover validation in places where social media POIs are not available.

Despite the advantages of social media POIs and the proposed approach, there is still much work that needs to be done to improve the proposed data and classification approach for land cover validation, which are as follows:

- (1) We use Weibo POIs as a representative dataset because of its popularity among the citizens of China and the large amount of available data that is relevant for our study. Considering the global scale of the GlobeLand30 data product, other social media data such as Twitter should also be used in the future to balance the uneven distribution of data. Furthermore, the text information in the geo-tagged photos can also be considered as a source of POI data which contains valuable information. Due to the varying popularity of social media applications in different countries, the use of improper datasets may lead to unsatisfactory results. In future studies, the sources of social media data should be broadened to help address the challenge of global land cover data product validation.
- (2) Artificial surfaces are set to be the validated land cover type in our study due to its vital importance in many applications. However, the validation of other land cover types, such as water bodies and cultivated land, are also necessary and can be utilized using social media POIs and the proposed classification method. For example, Wikipedia POIs provide useful information on water bodies, wetlands and other land cover types, which can be used in the proposed approach to support land cover validation. However, the spatial heterogeneity of other kinds of POI types is different from that of artificial surfaces, which makes the selection of modified decision tree parameters more difficult. This problem will be solved in our future work.
- (3) To classify non-urban POI data effectively, the shortest distances between POI and both villages and roads are considered as two parameters for identifying the potential non-artificial type of POIs. Although the result shows a 10% improvement compared to the unprocessed data, additional factors might also be incorporated into further experiments to improve the accuracy of the classification result. Moreover, apart from the classification process in non-urban areas, we should also take urban areas into consideration in subsequent studies, as the results in this experiment indicate a noticeable proportion of misidentified POIs.
- (4) The quality of POIs and other geo-tagged data directly affects land cover validation results. In this study, POIs are only pre-processed and evaluated based on the textual information because it is difficult to assess the quality of these volunteered contributing contents. However, the increasing emotional comments on these

POIs which indicate surrounding descriptions of POIs may help to evaluate the quality to some extent. Future studies will include POI dataset evaluation to assess their quality for more accurate validation.

6. Conclusions

Social media POI data contains valuable textual information about geo-location and descriptions of the surrounding environment, so it has great potential to be used for land cover validation. However, this kind of special textual data has seldom been analysed and used for supporting land cover validation. It is for this reason that the study described in this paper investigated a validation framework using modified decision trees to explore the usability of social media POI datasets for land cover validation. The proposed validation framework consists of two steps. First, POI classification was carried out through a modified decision tree model. To eliminate the effects of spatial heterogeneity on the classification of POIs, the shortest distances between each POI and both roads and villages served as two factors in the modified decision tree model. Second, classified POI data were transformed into a raster format based on a majority vote algorithm to agree with the spatial coverage and the resolution of the land cover map.

Experiments were conducted with 141,745 social media POIs from Sina Weibo covering the Beijing area to validate the artificial surface class of GlobeLand30-2010. A classification accuracy of 80.68% using Google Earth images as a reference was achieved through our modified decision tree method, while a classification accuracy of approximately 70% was obtained using a classification method that did not consider spatial heterogeneity. This indicates that our modified decision tree method has the ability to accurately classify POIs with high spatial heterogeneity. Using these classified POIs, a validation accuracy of 92.82% was achieved for the artificial surface class of GlobeLand30-2010, which is close to the validation accuracy of 86.7% derived in a formal validation by third-party experts. This proves that social media POI datasets can be used as one of the new reference sources for land cover validation, and the proposed validation framework has the ability to validate land cover maps with low costs in terms of money and time.

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